

Delta Neutral Positions and Dynamic Hedging

1. The process of delta hedging an option position to remain delta neutral is an important part of using a theoretical pricing model. In a delta neutral position all the deltas add up to approximately zero, or at least as close to zero as is practically possible. In this and subsequent exercises delta values are given in the whole number format.

	<u>July 70</u>	<u>July 80</u>	<u>July 90</u>
call delta	70	41	18
put delta	-30	-59	-82

- b.) You sell 50 July 70 puts. What do you need to do (buy? sell? how many?) to hedge your position as close to delta neutral as possible using each of the following contracts:

- i.) July 70 call

- ii.) July 80 put

- ii.) July 90 put

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	<u>July 70</u>	<u>July 80</u>	<u>July 90</u>
call delta	70	41	18
put delta	-30	-59	-82

- c.) You sell 15 underlying contracts. You would like to hedge half your delta position with July 80 puts and the other half with July 90 calls. As close as possible, how many of each contract do you need to buy or sell?

i) July 80 put

ii) July 90 call

Delta Neutral Positions and Dynamic Hedging

2. In each of the option positions below, how many underlying contracts do you need to buy or sell in order to create an initial delta neutral position. Then, as the delta changes, how many underlying contracts do you need to buy or sell in order to maintain a delta neutral position, or a position which is as close to delta neutral as possible? Delta values are given in the whole number format

a.) long 10 calls	<u>current option delta</u>	<u>current total delta position</u>	<u>underlying contracts to buy or sell</u>
	70	+700	
	50	-200	
	40	-100	
	80	+400	
b.) short 30 calls	<u>current option delta</u>	<u>current total delta position</u>	<u>underlying contracts to buy or sell</u>
	28	-840	
	40	-400	
	63	-690	
	52	+340	

Delta Neutral Positions and Dynamic Hedging

2. In each of the option positions below, how many underlying contracts do you need to buy or sell in order to create an initial delta neutral position. Then, as the delta changes, how many underlying contracts do you need to buy or sell in order to maintain a delta neutral position, or a position which is as close to delta neutral as possible? Delta values are given in the whole number format

c.) long 62 puts	<u>current option delta</u>	<u>current total delta position</u>	<u>underlying contracts to buy or sell</u>
	-65	-4030	
	-71	-402	
	-33	+2354	
	-49	-1038	
d.) short 44 puts	<u>current option delta</u>	<u>current total delta position</u>	<u>underlying contracts to buy or sell</u>
	-95	+4180	
	-77	-812	
	-40	-1640	
	0	-1600	

Delta Neutral Positions and Dynamic Hedging

3. In theory an option's value should be equal to the present value of its intrinsic value at expiration, plus the present value of all the cash flows resulting from the delta neutral dynamic hedging process.

In the two scenarios below, assume that the option is hedged at each given underlying price and delta. If interest rates are zero (no present valuing is necessary), what are the values of each option?

a.)	<u>option</u>	<u>underlying price</u>	<u>delta</u>	
	55 call	53.70	45	(initial underlying price and delta)
		55.70	57	
		53.40	40	
		57.10	69	
		55.00	51	
		51.50		(underlying price at expiration)

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b.)	<u>option</u>	<u>underlying price</u>	<u>delta</u>	
	70 put	68.42	-65	(initial underlying price and delta)
		69.71	-52	
		71.29	-33	
		70.19	-46	
		71.02	-28	
		68.51		(underlying price at expiration)

Delta Neutral Positions and Dynamic Hedging (answers)

1. The process of delta hedging an option position to remain delta neutral is an important part of using a theoretical pricing model. In a delta neutral position all the deltas add up to approximately zero, or at least as close to zero as is practically possible. In this and subsequent exercises delta values are given in the whole number format.

	<u>July 70</u>	<u>July 80</u>	<u>July 90</u>
call delta	70	41	18
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- a.) You buy 25 July 80 calls. From the table of delta values above, what do you need to do (buy? sell? how many?) to hedge your position as close to delta neutral as possible using each of the following contracts:

$$\text{delta position} = 25 \times 41 = \mathbf{+1025}$$

- | | | |
|-------------------------|------------------------|----------------|
| i.) underlying contract | $+1025 / 100 = 10.25$ | sell 10 |
| ii.) July 90 call | $+1025 / 18 = 56.94$ | sell 57 |
| ii.) July 80 put | $+1025 / -59 = -17.37$ | buy 17 |

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	<u>July 70</u>	<u>July 80</u>	<u>July 90</u>
call delta	70	41	18
put delta	-30	-59	-82

- b.) You sell 50 July 70 puts. What do you need to do (buy? sell? how many?) to hedge your position as close to delta neutral as possible using each of the following contracts:

$$\text{delta position} = -50 \times -30 = \mathbf{+1500}$$

- | | | |
|------------------|------------------------|----------------|
| i.) July 70 call | $+1500 / 70 = 21.43$ | sell 21 |
| ii.) July 80 put | $+1500 / -59 = -25.42$ | buy 25 |
| ii.) July 90 put | $+1500 / -82 = -18.29$ | buy 18 |

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	<u>July 70</u>	<u>July 80</u>	<u>July 90</u>
call delta	70	41	18
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- c.) You sell 15 underlying contracts. You would like to hedge half your delta position with July 80 puts and the other half with July 90 calls. As close as possible, how many of each contract do you need to buy or sell?

$$\text{delta position} = -15 \times 100 = \textbf{-1500} \quad -1500 / 2 = -750$$

$$\text{July 80 put} \quad -750 / -59 = +12.71 \quad \textbf{sell 13}$$

$$\text{July 90 call} \quad -750 / 18 = -41.67 \quad \textbf{buy 42}$$

Delta Neutral Positions and Dynamic Hedging (answers)

2. In each of the option positions below, how many underlying contracts do you need to buy or sell in order to create an initial delta neutral position. Then, as the delta changes, how many underlying contracts do you need to buy or sell in order to maintain a delta neutral position, or a position which is as close to delta neutral as possible? Delta values are given in the whole number format

a.) long 10 calls	<u>current option delta</u>	<u>current total delta position</u>	<u>underlying contracts to buy or sell</u>
	70	+700	<i>sell 7</i>
	50	-200	<i>buy 2</i>
	40	-100	<i>buy 1</i>
	80	+400	<i>sell 4</i>
b.) short 30 calls	<u>current option delta</u>	<u>current total delta position</u>	<u>underlying contracts to buy or sell</u>
	28	-840	<i>buy 8</i>
	40	-400	<i>buy 4</i>
	63	-690	<i>buy 7</i>
	52	+340	<i>sell 3</i>

2. In each of the option positions below, how many underlying contracts do you need to buy or sell in order to create an initial delta neutral position. Then, as the delta changes, how many underlying contracts do you need to buy or sell in order to maintain a delta neutral position, or a position which is as close to delta neutral as possible? Delta values are given in the whole number format

c.) long 62 puts	<u>current option delta</u>	<u>current total delta position</u>	<u>underlying contracts to buy or sell</u>
	-65	-4030	buy 40
	-71	-402	buy 4
	-33	+2354	sell 24
	-49	-1038	buy 10

d.) short 44 puts	<u>current option delta</u>	<u>current total delta position</u>	<u>underlying contracts to buy or sell</u>
	-95	+4180	sell 42
	-77	-812	buy 8
	-40	-1640	buy 16
	0	-1600	buy 18

3. In theory an option's value should be equal to the present value of its intrinsic value at expiration, plus the present value of all the cash flows resulting from the delta neutral dynamic hedging process.

In the two scenarios below, assume that the option is hedged at each given underlying price and delta. If interest rates are zero (no present valuing is necessary), what are the values of each option?

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		53.40	40	
		57.10	69	
		55.00	51	
		51.50		(underlying price at expiration)

3) a

<u>underlying price</u>	<u>delta</u>	<u>hedge</u>	<u>cash flow</u>	<u>total hedge</u>
53.70	45	sell .45	$+.45 \times 53.70 = +24.17$	-.45
55.70	57	sell .12	$+.12 \times 55.70 = +6.68$	-.57
53.40	40	buy .17	$-.17 \times 53.40 = -9.08$	-.40
57.10	69	sell .29	$+.29 \times 57.10 = +16.56$	-.69
55.00	51	buy .18	$-.18 \times 55.00 = -9.90$	-.51
51.50		buy .51	$-.51 \times 51.50 = -26.27$	0

Dynamic hedging cash flow: **$+24.17 + 6.68 - 9.08 + 16.56 - 9.90 - 26.27 = +2.16$**

Intrinsic value of the 55 call at expiration: 0

Theoretical value of the 66 call: $2.16 + 0 = \underline{2.16}$

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		71.29	-33	
		70.19	-46	
		71.02	-28	
		68.51		(underlying price at expiration)

3) b

<u>underlying price</u>	<u>delta</u>	<u>hedge</u>	<u>cash flow</u>	<u>total hedge</u>
68.42	-65	buy .65	$-.65 \times 68.42 = -44.47$	+.65
69.71	-52	sell .13	$+.13 \times 69.71 = +9.06$	+.52
71.29	-33	sell .19	$+.19 \times 71.29 = +13.55$	+.33
70.19	-46	buy .13	$+.13 \times 70.19 = -9.12$	+.46
71.02	-28	sell .18	$+.18 \times 71.02 = +12.78$	+.28
68.51		sell .28	$+.28 \times 68.51 = +19.18$	0

Dynamic hedging cash flow: $-44.47 + 9.06 + 13.55 - 9.12 + 12.78 + 19.18 = +.98$

Intrinsic value of the 70 put at expiration: $70.00 - 68.51 = 1.49$

Theoretical value of the 66 call: $.98 + 1.49 = \underline{2.47}$